
CASE PRESENTATION ON CHRONIC CALCULUS CHOLECYSTITIS

PRESENTED BY

AASHRITHA.S

23Z11T0001

DEFINITION

- Chronic calculus cholecystitis is a chronic inflammation of the gall bladder caused by the presence of gall stones usually resulting from repeated episodes of gall bladder irritation and inflammation
- It is one of the most common chronic disorders of the gall bladder
- Gall stones repeatedly irritates the gall bladder and may cause intermittent obstruction of cystic duct.
- Persistent inflammation leads to thickening, fibrosis and reduced contractility of gall bladder

ETIOLOGY

- Gall stones [cholethiasis] – the main cause
- Repeated cystic duct obstruction by gall stones
- Repeated episodes of acute cholecystitis leads to chronic inflammation
- Bile stasis and impaired gallbladder emptying
- Bacterial infection[E.coli, klebsiella, enterobacter etc.] may contribute in some cases

RISK FACTORS

Female sex

Increasing age

Obesity

High fat diet

Rapid weight loss

Diabetes mellitus

Family history of gall stones

SIGNS AND SYMPTOMS

- Right upper abdominal pain
- Pain radiates to right shoulder or back
- Dyspepsia
- Bloating, indigestion
- Occasionally fever due to acute exacerbation
- Positive murphys sign
- Pain often occurs after fatty meals

DIAGNOSIS



History and physical examination



Ultrasonography of abdomen and pelvis



Complete blood picture



HIDA scan [cholecintigraphy]



CT, MRI scan



Liver and renal function tests

SUBJECTIVE DATA

Patient name: xxx

Age: 76years

Gender: female

Department: general surgical ward

IP NO: 260634523

Date of admission: 24/6/26

Date of discharge: 1/7/26

INITIAL ASSESSMENT

- **CHIEF COMPLAINTS:**
- Complaint of pain in abdomen since 2 months no significant associated bleeding, nausea/ vomiting or bowel or bladder complaints.
- **HISTORY OF PRESENT ILLNESS:**
- Patient was apparently alright 2 months back then to start with she developed pain in abdomen which was insidious in onset, gradually progressive to present condition, no aggravating or relieving factors observed

INITIAL ASSESSMENT...

- HISTORY OF PAST ILLNESS:
 - K/C/O Hypertension since 6 years on regular medication Tab. Telmisatran[20mg/PO/ OD]
- PAST MEDICAL/ SURGICAL HISTORY :
 - H/O tubectomy 40years back
- FAMILY HISTORY: NAD

PERSONAL HISTORY:

- Height: 165cms
- Weight : 90kgs
- Alcohol: 3 times per week 90ml [last intake- 2 days back]
- No habit of smoking
- Mixed diet
- Bowel and bladder: regular
- Appetite: normal
- Sleep : adequate

PHYSICAL EXAMINATION

GCS: fair

BP:
130/80mmHg

Temperature:
98.5 F

Lungs: BAE+

CVS: S1S2+

PR: 88Bpm

RR: 16cpm

Spo2:
99%@RA

LOCAL EXAMINATION

- **Inspection-** abdomen appears slightly distended, E/O a visible mid line vertical scar of 6*1cm noted extending from 3cm below umbilicus and above pubic symphysis
- E/O 3*2 cm scar noted 2cm above umbilicus
- **Percussion-** liver dullness noted

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- PROVISIONAL DIAGNOSIS



PAIN IN ABDOMEN
CHOLECYSTITIS



EVALUATION

OBJECTIVE DATA

COMPLETE BLOOD COUNT

	DAY 1	DAY 2	NORMAL RANGE
TOTAL WBC COUNT	10500	9100	4000-10000 cells/cu.mm
NEUTROPHILS	66%	63%	40-80%
HAEMOGLOBIN	11.6g/dL	11.6g/dL	12-15g/ dL
HEMATOCRIT	37.7%	36.9%	38-51% pcv
MCV/MCH	91.1fL /28.8pg	90.9fL / 28.5pg	83-101fL/ 27-32pg
MCHC	31.6g/dL	31.4g/ dL	31.5- 34.5g/ dL

RENAL FUNCTION TEST

	EXAMINED VALUE	NORMAL RANGE
SERUM CREATININE	0.8mg/dL	0.6-1.1mg/dL
SODIUM	138mmol/L	135-155mmol/L
POTASSIUM	4.4mmol/L	3.5-5.5mmol/L

LIVER FUNCTION TEST

	EXAMINED VALUE	NORMAL RANGE
TOTAL BILIRUBIN	0.60mg/dL	0.3-1.2 mg/dL
DIRECT BILIRUBIN	0.20mg/dL	0.1-0.4 mg/dL
SERUM TOTAL PROTEIN	6.2g/dL	6-8g/dL
ALBUMIN	4.2g/dL	3.5-5.2g/dL
GLOBULIN	2.0g/dL	2.6-4.6g/dL
A/G RATIO	2.1	1.2-1.5
ALP	83U/L	53-128U/L

OTHER FINDINGS

ULTRASONOGRAPHY OF ABDOMEN AND PELVIS

Gall bladder: contracted

Impression:

- grade 1 fatty liver
- Renal parenchymal change

UPPER GI ENDOSCOPY:

- Esophagus: mucosal breaks < 5cm in distal esophagus
- Stomach: erosions in body and antrum
- Impression: LA grade A esophagitis with a hypotensive lower esophageal sphincter involving erosive gastritis.

MAGNETIC RESONANCE CHOLANGIOPANCREATOGRAPHY



Gallbladder was contracted with mildly thickened wall and no internal sludge
no pericholecystic inflammatory changes



Impression: findings suggestive of chronic cholecystitis

SOAP ANALYSIS

SUBJECTIVE DATA: A 76year old female patient was admitted in the female surgical ward with chief complaints of pain in abdomen which was insidious in onset, gradually progressive to present condition, no aggreveating or relieving factors are observed

OBJECTIVE DATA WBC COUNT-9100Cells/cu.mm

INVESTIGATIONS: CBP reveals low hemoglobin [11.6g/dL] and LFT revealed low levels of globulin [2.0g/dL] indicates hypoglobulinemia

IMAGING FINDINGS: USG findings suggestive of cholethiasis with grade 1 fatty liver

MRCP: contracted gallbladder with mildly thickened wall and internal sludge CBD is mildly dilated[10mm]

ENDOSCOPY: LA Grade A esophagitis, hypotensive LES and erosive gastritis

- **ASSESSMENT:** Based on subjective and objective evidence this case is assessed as
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CHRONIC CALCULUS

CHOLECYSTITIS

PLAN OF CARE

Patient with chronic calculus cholecystitis and associated gallbladder disease



Confirm and monitor the investigations [CBP, LFTs, serum electrolytes, urine analysis, MRCP/USG]



Assess the severity of symptoms[abdominal pain, pain severity score, abdominal tenderness,



PAC



Pre-operative stabilization: IV fluids for dehydration

- analgesics for pain relief
- anti emetics for nausea and vomiting



Surgical management: laparoscopic cholecystectomy



Post operative care and patient counselling

THERAPEUTIC GOALS

- Relieve abdominal pain and discomfort
- Control inflammation and prevent complications of chronic cholecystitis
- Treat gallstone disease and prevent recurrent biliary symptoms
- Definitive treatment with laparoscopic cholecystectomy to remove diseased gallbladder
- Manage nutritional abnormalities, including mild anemia and low globulin, as clinically appropriate

TREATMENT CHART

S. N O	BRAND NAME	ACTIVE COMPOSIT ION	INDICATIO N	DOSE	RO A	FRE QUE NCY	SIDE EFFECTS	START DATE	STOP DATE
1	Tab.pan	pantoprazol e	Proton pump inhibitor	40mg	PO	OD	headache Vit-b12 deficiency	25/6/26	28/6/26
2	Tab.buscopan	Hyoscine butylbromid e	Anti cholinergic	10mg	PO	SOS	Dry mouth constipation	25/6/26	28/6/26
3	Tab.telma	telmisatran	ARB	20mg	PO	OD	Postural hypotension	25/6/26	28/6/26
4	Inj.magne x forte	Cefperazon e+ sulbactam	Cephalspor in antibiotic	1.5g m	IV	12 th hrly	Inj.site rxns nausea	27/6/26	29/6/26
5	Inj. metrogyl	metronidazo le	Nitroimidaz ole antibiotic	500m g	IV	8 th hrly	Metallic taste, dizziness	27/6/26	29/6/26

S.NO	BRAND NAME	ACTIVE COMPOSITION	INDICATION	DOSE	ROUTE	FREQUENCY	SIDE EFFECTS	START DATE	STOP DATE
6	Inj. tramadol	tramadol	Opioid analgesic	1amp in 100ml NS	IV	12 th hrly	Drowsiness, headache	27/6/26	29/6/26
7.	Inj. zofer	ondansetron	Antiemetic	4mg	IV	12 th hrly	Constipation headache	27/6/26	29/6/26
8.	Tab.limcee	Ascorbic acid	Vitamin supplement	500mg	PO	OD	Abdominal cramps diarrhoea	28/6/26	30/6/26

9	Tab havit	Cholecalciferol[vit-D3]	Vitamin D supplement	60,000IU	PO	OD	Constipation, hypercalcaemia[in high doses]	28/6/26	30/6/26
10	Tab.medpocv	Cefpodoxime proxetil + clavulanic acid	Cephalosporin antibiotic	325mg	PO	BD	Rash, allergic rxns	29/6/26	30/6/26
11	Syp.duphalac	lactulose	Osmotic laxative	30ml	PO	HS	Diarrhoea, flatulence	30/6/26	1/7/26
12	Tab.trypace	Aceclofenac + paracetamol+ trypsin+ chymotrypsin	NSAID+ Antipyretic+ proteolytic enzyme	1tablet	PO	BD	Gastric irritation, abdominal pain	29/6/26	30/6/26

PHARMACIST INTERVENTION

Major drug-drug and drug- food interactions were found in the prescription

1. Tramadol+ ondansetron- ondansetron reduces the analgesic effect of tramadol by interfering with serotogenic pain pathways
2. Telmisatran+ dynapar patch- dynapar reduces the antihypertensive effect and increases the risk of kidney injury
3. Metronidazole+ ondansetron- both the drugs increases QT prolongation, especially in elder patients. The alternative medication is metoclopramide
4. Tramadol + alcohol- alcohol increases the CNS and respiratory depression. Avoid alcohol completely during tramadol therapy
5. Telmisatran+ high potassium foods[potato, banana, avocado etc]- results in hyperkalemia

PROGRESSIVE CHART

Date	GC	TEMP	BP	COMPLAINTS	ADDED DRUGS	REMOVED DRUGS
25/6/26	Fair	97F	110/70mmHg	-----	T.Pan Tab.buscopanTa b. telma	-----
26/6/26	Fair	98.2F	160/90mmHg	-----	c/s/t	-----
27/6/26	Fair	98.3F	140/90mmHg	-----	Inj.magnex forte Inj.metrogyl Inj.tramadol Inj.zofer	-----
28/6/26	fair	98 F	130/90mmHg	-----	Tab limcee Tab.havit	-----
29/6/26	fair	97F	140/90mmHg	-----	Tab.medpod-cv, Tab.trypace	-----
30/6/26	fair	98.2F	120/90mmHg	-----	Syp.duphlalac	-----

SURGICAL PROCEDURE



LAPAROSCOPIC CHOLECYSTECTOMY WAS PERFORMED UNDER GENERAL ANAESTHESIA



Procedure: pneumoperitoneum was created and ports inserted. Gallbladder was dissected and cystic duct, artery were clipped and divided. Gallbladder was separated from liver bed and removed

DISCHARGE STATUS: CURED DISCHARGE SUMMARY

- Patient name: xxx
- Final diagnosis: chronic calculus cholecystitis
- Vitals:
 - temp: 98.2F
 - PR: 78bpm
 - BP: 160/80mmHg
 - RR: 16cpm
 - spo2: 98%@RA

DISCHARGE MEDICATIONS

DRUG	DOSE	ROA	FREQUENCY
TAB.MEDPOD-CV	325mg	PO	BD
TAB.PAN	40mg	PO	OD
TAB.TRYPACE	1tab	PO	BD
TAB.LIMCEE	500mg	PO	BD

PATIENT COUNSELLING REGARDING DIET AND LIFESTYLE MODIFICATIONS

- Drink adequate water and maintain hydration
- Avoid alcohol, especially during the recovery period
- Avoid heavy meals, take small, frequent meals
- Keep the surgical wound clean and dry and follow wound care instructions
- Take all the prescribed medicines at correct dose and time
- Walking is encouraged as tolerated to improve mobility and prevent constipation
- Avoid high lifting and strenuous exercise until the surgeon advises that it is safe

PATIENT COUNSELLING REGARDING THE MEDICINES

1. Tab.medpod-cv: it consist of cefpodoxime and clavulanic acid. An antibiotic which is used to prevent bacterial infection after surgery. Donot skip the dose. show side effects like abdominal pain, headache etc.
2. Tab. PAN: it consists of pantoprazole. It reduces acidity and gastric irritation . Take daily preferably 30mins before breakfast. Show adrs like diarrhoea, vit b12 deficiency etc..
3. Tab.trypace: it consists of aceclofenac, paracetamol, trypsin+ chymotrypsin. Used to reduce post operative swelling and inflammation. Avoid additional NSAIDS, painkillers. Show side effects like indigestion, stomach pain
4. Tab.limcee: vitamin supplementation which consists of ascorbic acid. Acts as anti-oxidant, and support wound healing. Chew the tablet completely before swallowing. Show side effects like nausea, GI upset etc..